

Time:(3 Hours)

marks:80

N.B: 1) Question **number 1** is compulsory.

2) Attempt **any three** out of the remaining.

3) Assume suitable data if **necessary** and justify the assumptions.

4) Figures to the **right** indicate full marks.

Q 1

- A Explain how Monotonic read consistency model is different than Read your Write consistency model [5]
- B What is 1:M and M:1 group communication? [5]
- C Differentiate between NOS DOS and Middleware in the design of distributed systems [5]
- D What is fault tolerance? Explain failure models. [5]

Q 2

- A Explain code migration and its techniques. [10]
- B Explain Bully election algorithm with suitable example. [10]

Q 3

- A Explain Raymond's algorithm for mutual exclusion. [10]
- B What are different data centric consistency models [10]

Q 4

- A Explain different load estimation and process transfer policies used by load balancing algorithms. [10]
- B What are physical clocks? Explain any one physical clock synchronization algorithm. [10]

Q 5

- A What are different issues and goals related to distributed systems? [10]
- B Explain file caching schemes [10]

Q 6

- A Write a short note on Replication and the types of it. [10]
- B What is RPC? Explain working of RPC in detail [10]

Duration: 3hrs

[Max Marks:80]

- N.B. : (1) Question No 1 is Compulsory.
 (2) Attempt any three questions out of the remaining five.
 (3) All questions carry equal marks.
 (4) Assume suitable data, if required and state it clearly.

- 1 Attempt any FOUR [20]
 - a Explain issues in designing Distributed system
 - b Compare NOS and DOS
 - c Explain desirable features of global scheduling algorithm
 - d Explain the need of election algorithm.
 - e Justify how Ricart-Agrawala's algorithm optimized the Message overhead in achieving mutual exclusion
- 2 a What is Remote procedure call? Explain how transparency is achieved in RPC [10]
 - b Explain various forms of message oriented communication with suitable example [10]
- 3 a What is logical clock? Why are logical clocks required in distributed systems? How Lamport does synchronizes logical clock? Which events are said to be concurrent in Lamports timestamp [10]
 - b Explain Chandy -Misra_Hass Algorithm for distributed deadlock detection. [10]
- 4 a Explain different load estimation and process transfer policies used by load balancing algorithms. [10]
 - b Describe code migration issues in details [10]
- 5 a Discuss and differentiate various client consistency models. [10]
 - b Explain Absolute ordering and Casual ordering process with the help of example for many to many communication. [10]
- 6 a List desirable features of distributed File system. How are modifications propagated in file caching schemes? [10]
 - b Discuss Raymonds tree based algorithm of token based in distributed mutual exclusion [10]

(Time: 3 Hours)

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2) Attempt **any THREE** out of the remaining.
3) Assume suitable data if **necessary** and justify the assumptions.
4) Figures to the **right** indicate full marks.

Q 1

- A Explain key Features of Global Scheduling algorithm. [05]
B What are the services offered by middleware? [05]
C Differentiate between RMI and RPC. [05]
D Explain synchronization in DFS with its challenges. [05]

Q 2

- A Justify how load balancing is useful in distributed system. [10]
B What is fault tolerance? Explain failure models. [10]

Q 3

- A Explain group communication. [10]
B Explain Raymond's tree based algorithm for mutual exclusion. [10]

Q 4

- A Explain Ricart-Agrawala's algorithm and how it optimizes the message overhead in achieving mutual exclusion. [10]
B Discuss Google file system (GFS) as a scalable distributed file system. [10]

Q 5

- A Explain bully election algorithm. [10]
B Discuss code migration in distributed system. [10]

Q 6

- A Explain any five data centric consistency models. [10]
B Discuss design issues in distributed systems. [10]

BE(comp) R19 c scheme 13-05-25

(Time: 3 Hours)



Marks: 80

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 2) Attempt **any three** out of the remaining.
 3) Assume suitable data if **necessary** and justify the assumptions.
 4) Figures to the **right** indicate full marks.

Q 1

- A Enlist the issue in designing the Distributed System? Explain failure transparency & location transparency in detail [5]
 B What are the key features of Global Scheduling algorithm. [5]
 C Explain Bully algorithm with example [5]
 D Explain the various ordered semantics used for Many to Many communication [5]

Q 2

- A Explain Chandy Misra Hass Algorithm [10]
 B Desirable features of a good DFS? [10]

Q 3

- A Explain load estimation, process transfer and location policies with respect to load balancing approach in distributed systems. [10]
 B Explain Raymond's algorithm for mutual exclusion. [10]

Q 4

- A Explain the goals of distributed systems. [10]
 B How Lamport does synchronizes logical clock explain with example? Which events are said to be concurrent in Lamport's timestamp. [10]

Q 5

- A What is fault tolerance? Describe different types of failure models. [10]
 B What is RPC? Explain working of RPC in detail [10]

Q 6

- A Discuss the technique to achieve the Process resilience. [10]
 B What is need of code migration? Explain the role of process to resource and resource to machine binding in code migration. [10]

Op code

82386

Prog code
1T00738